

IS4103

Operating Systems

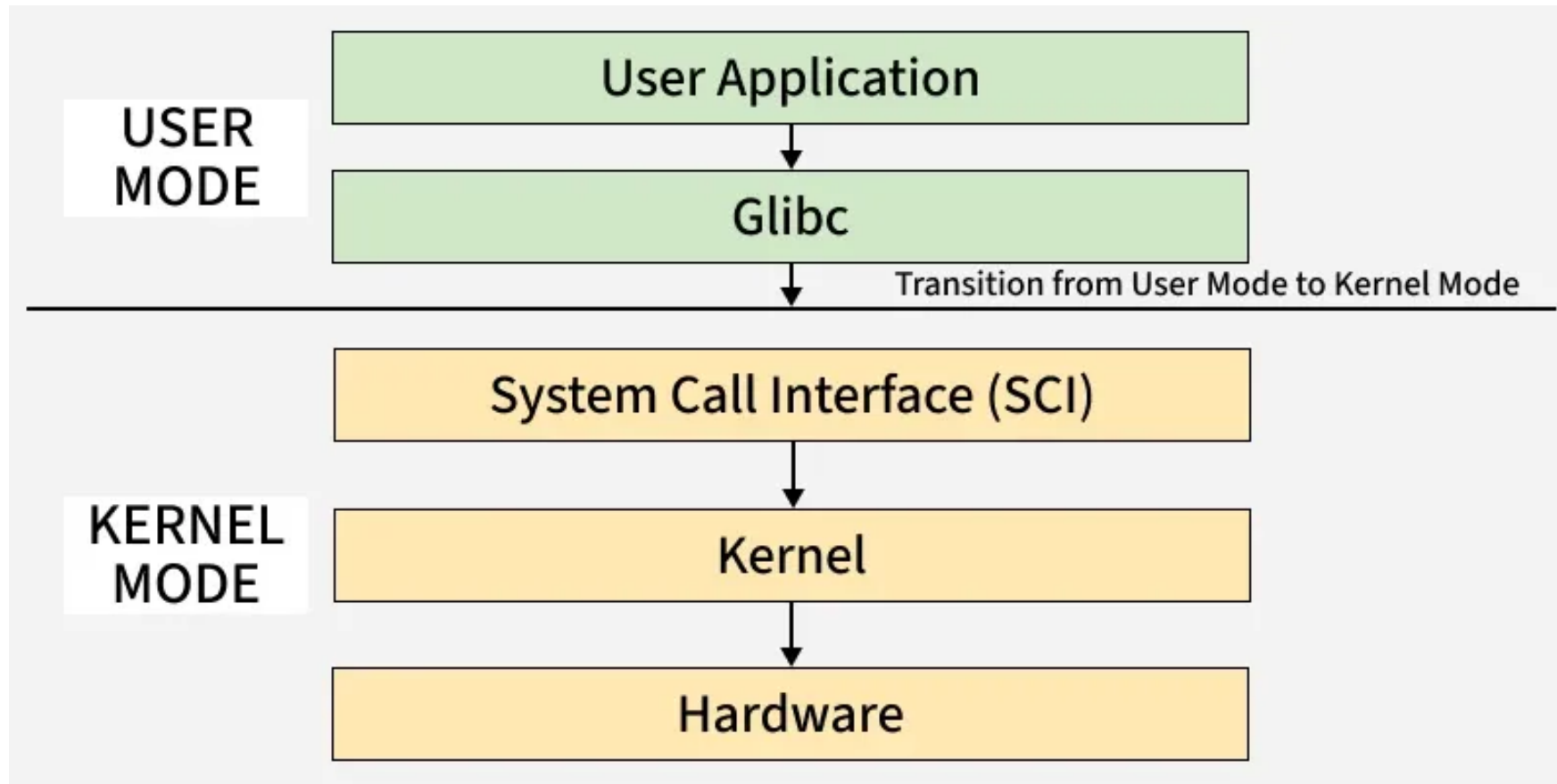
Operating Systems Architecture

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Learning Objectives

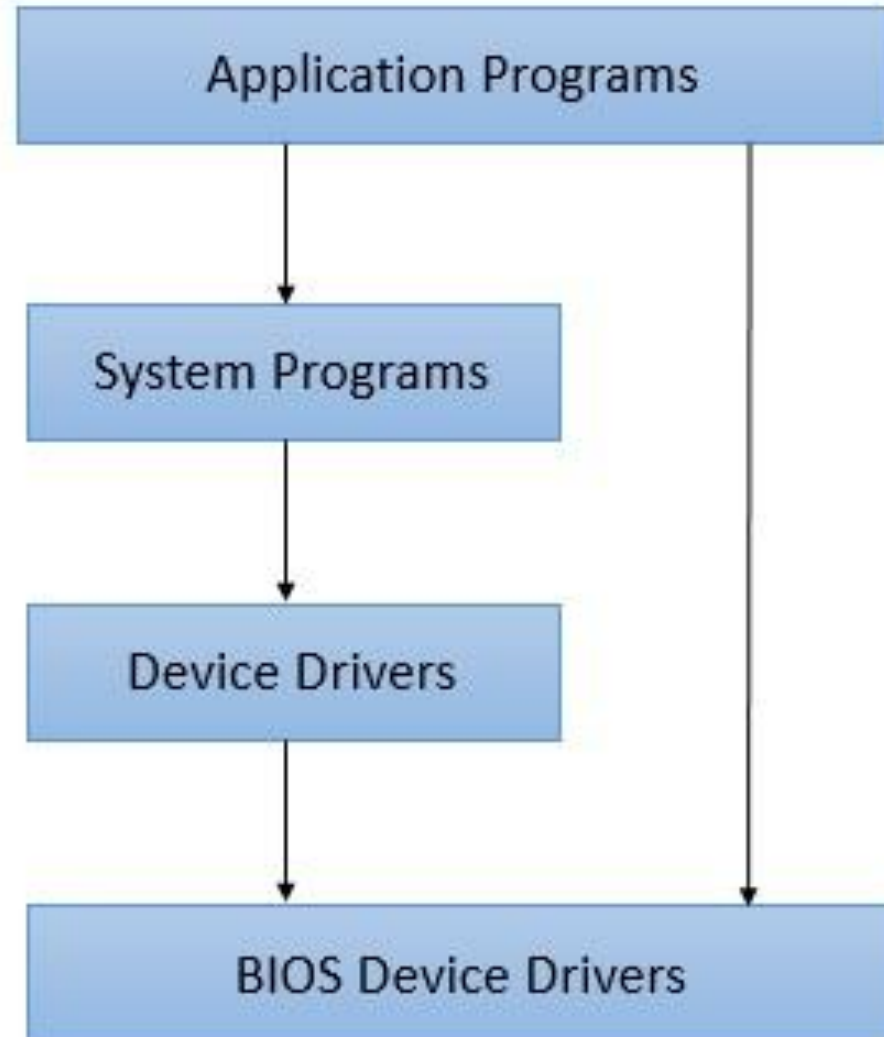
Learning Outcome	Description	At the end of Today's Session
ILO1	Describe the including development & achievements, functionalities & objectives, structure & components	• Types of Architectures in Operating Systems

OS Models



Basis of difference	User mode	Kernel mode
Modes	User Mode is regarded as the restricted mode or the slave mode.	Kernel Mode is considered the master mode, system mode or the privileged mode.
Definition	In restricted User Mode, the application programs execute and start.	The computer enters the Kernel mode, which is a privileged mode while accessing hardware resources.
Interruptions	In case any interrupt occurs in User Mode, only one process fails.	In case an interrupt occurs in Kernel Mode, the complete operating system might fail.
Restrictions	There are restrictions in user mode to access kernel programs. They cannot be accessed directly.	Both kernel programs, as well as user programs, can access in kernel mode.

Simple OS Architecture



Monolithic OS Architecture



Applications

User Mode

handles other OS features

Kernel Mode

handles only basic OS features

File system
Management

Process
Management

Memory
Management

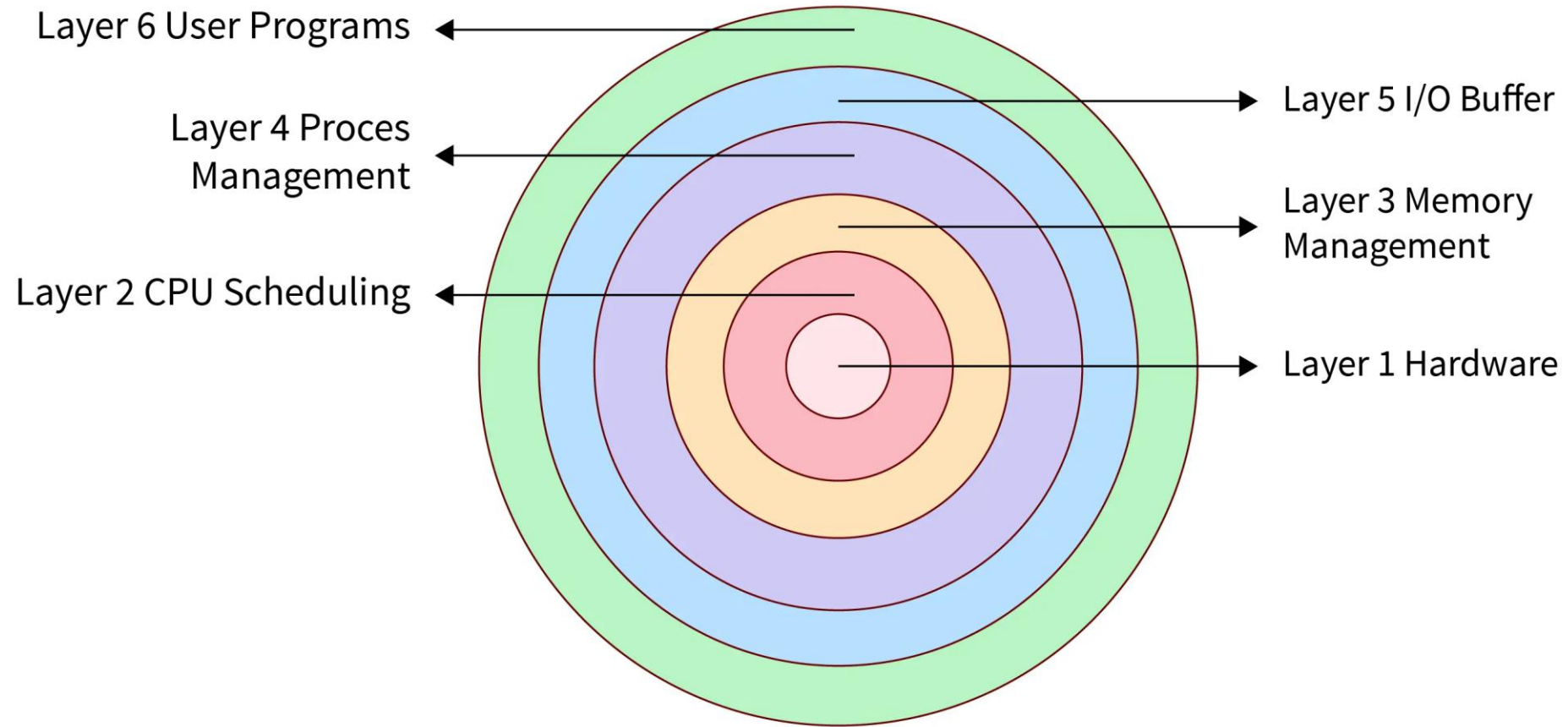
I/O
Management

Concurrency

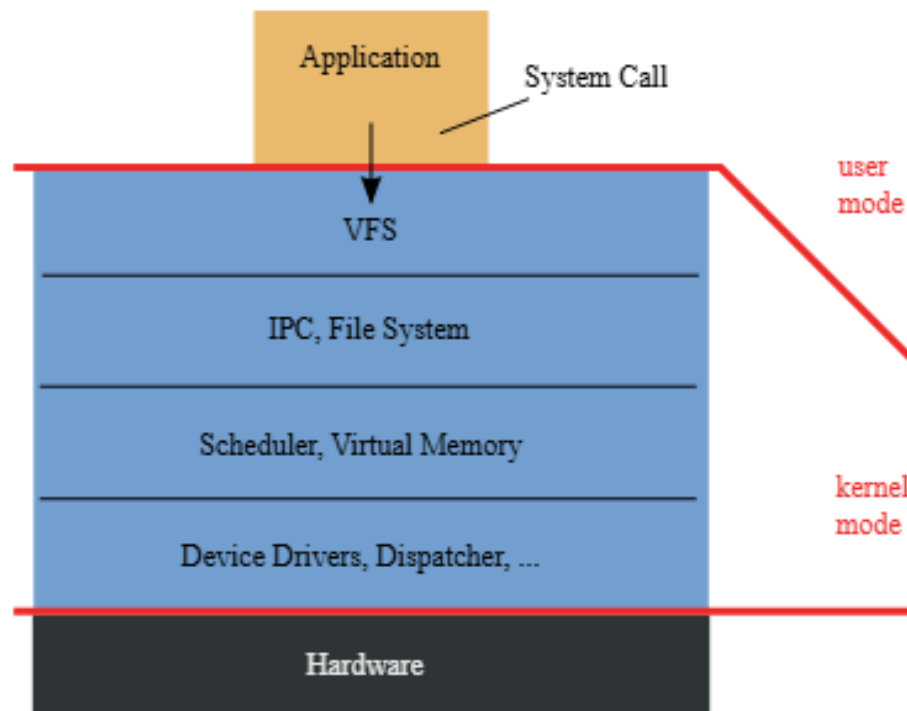
Device
management

Hardware

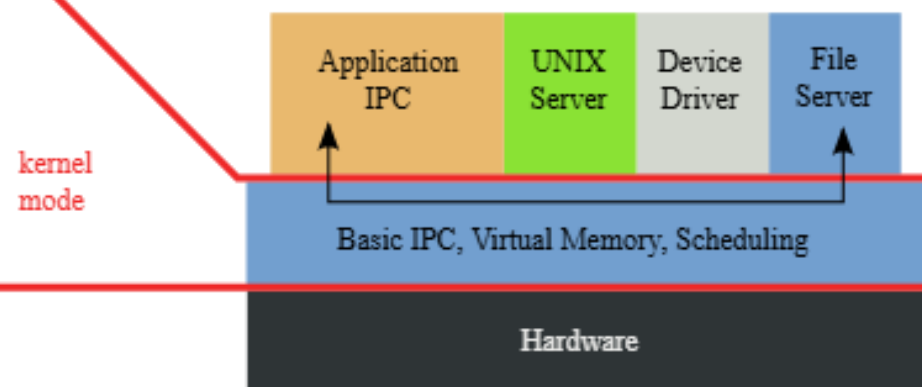
Layered OS Architecture



Monolithic Kernel based Operating System



Microkernel based Operating System



OS Hybrid Architecture

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Applications

User Mode

handles other OS features

Process
Management

File
Servers

I/O
Management

Memory
Management

Kernel Mode

handles only basic OS features

Device
Drivers

Low Level
Address space
Management

Low Level
process
Management

Inter process
Communication

Hardware

Thank you

Any Questions?